zero here. But, they have equal and opposite effects, but we assume the loop to be rigid, so no deformation. Hence, no torque produced by these two sections.

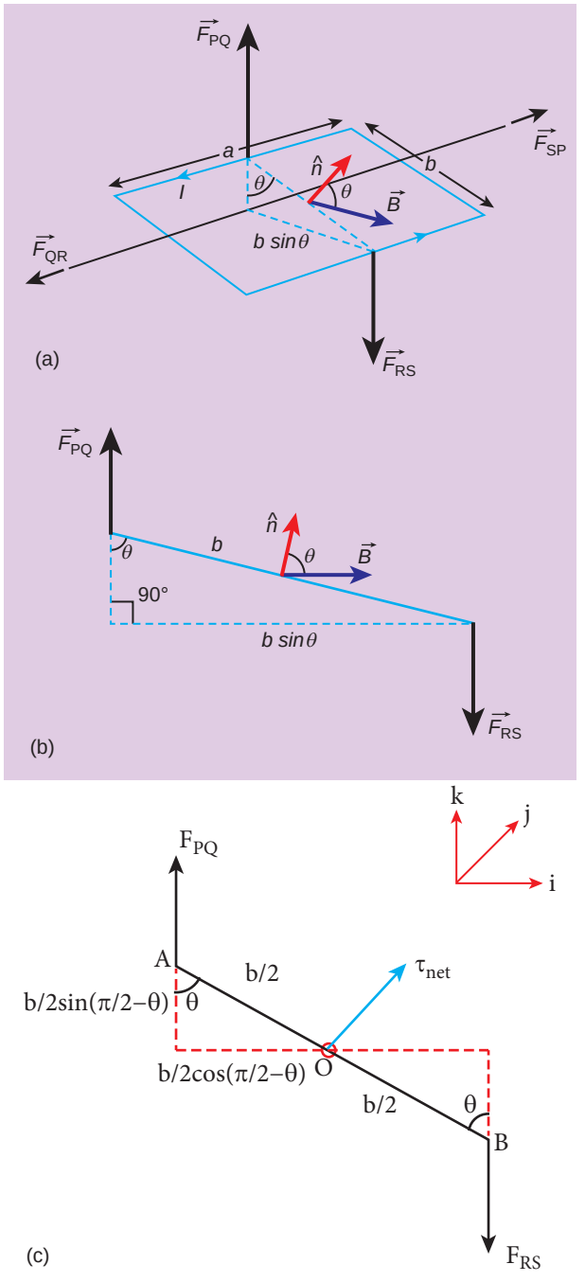


Figure 3.65 Force on the rectangular loop – (a) top view and (b) side view (c) net torque on the loop

Even though the forces PQ and RS also are also equal and opposite, they are not collinear. So these two forces constitute a couple as shown in Figure 3.65 (a). Hence the net torque produced by these two forces about the axis of the rectangular loop is given by

$$\vec{\tau}_{net} = baBI\sin\theta\,\hat{k} = ABI\sin\theta\,\hat{k}$$

From the Figure 3.65 (c),

$$\overrightarrow{OA} = \frac{b}{2}\cos\left(\frac{\pi}{2}-\theta\right)(-\hat{i}) + \frac{b}{2}\sin\left(\frac{\pi}{2}-\theta\right)(-\hat{k})$$

$$= \frac{b}{2}(-\sin\theta\,\hat{i} + \cos\theta\,\hat{k})$$

$$\overrightarrow{OB} = \frac{b}{2}\cos\left(\frac{\pi}{2}-\theta\right)(\hat{i}) + \frac{b}{2}\sin\left(\frac{\pi}{2}-\theta\right)(-\hat{k})$$

$$= \frac{b}{2}(-\sin\theta\,\hat{i} + \cos\theta\,\hat{k})$$

$$\overrightarrow{OA}\times\vec{F}_{PQ} = \left\{\frac{b}{2}(-\sin\theta\,\hat{i} + \cos\theta\,\hat{k})\right\}\times\{IaB\,\hat{k}\}$$

$$= \frac{1}{2}IabB\sin\theta\,\hat{j}$$

$$\overrightarrow{OB}\times\vec{F}_{RS} = \left\{\frac{b}{2}(\sin\theta\,\hat{i} + \cos\theta\,\hat{k})\right\}\times\{-IaB\,\hat{k}\}$$

$$= \frac{1}{2}IabB\sin\theta\,\hat{j}$$

The net torque $\vec{\tau}_{net} = IabB\sin\theta\,\hat{j}$ (3.68)

Note that the net torque is in the positive y direction which tends to rotate the loop in clockwise direction about the y axis. If the current is passed in the other way (P→S→R→Q→P), then total torque will point in the negative y direction which tends to rotate the loop in anticlockwise direction about y axis.

Another important point is to note that the torque is less in this case compared to earlier case (where the $\hat{n}$ is perpendicular to the magnetic field $\vec{B}$). It is because the perpendicular distance is reduced between the forces $\vec{F}_{PQ}$ and $\vec{F}_{RS}$ in this case.

The equation (3.68) can also be rewritten in terms of magnetic dipole moment $\vec{p}_m = I\vec{A} = Iab\,\hat{n}$

$$\vec{\tau}_{net} = \vec{p}\times\vec{B}$$

UNIT-3(XII-Physics_Vol-1).indd 189 04-03-2019 10:35:45